Setting Up a Typical Campus Network

Difficulty (out of 5 stars): ★★★★★ Recommended study time: 4 hours

Contents

About This Lab

Task Description

Campus networks are the key infrastructure that connects campuses to the digital world, making them an indispensable part of campus construction. They are playing an increasingly important role in daily work, R&D, production, and operation management.

In this lab activity, you will set up a campus network to understand common technologies and their applications on campus networks. After completing this task, you will have a clear understanding of campus network construction and be familiar with the construction and configuration of campus networks.

Objectives

On completing this task, you will be able to:

1. Understand common campus network concepts and architecture.
2. Understand common network technologies.
3. Understand campus network deployment and implementation.

Task Preparation

Network Topology

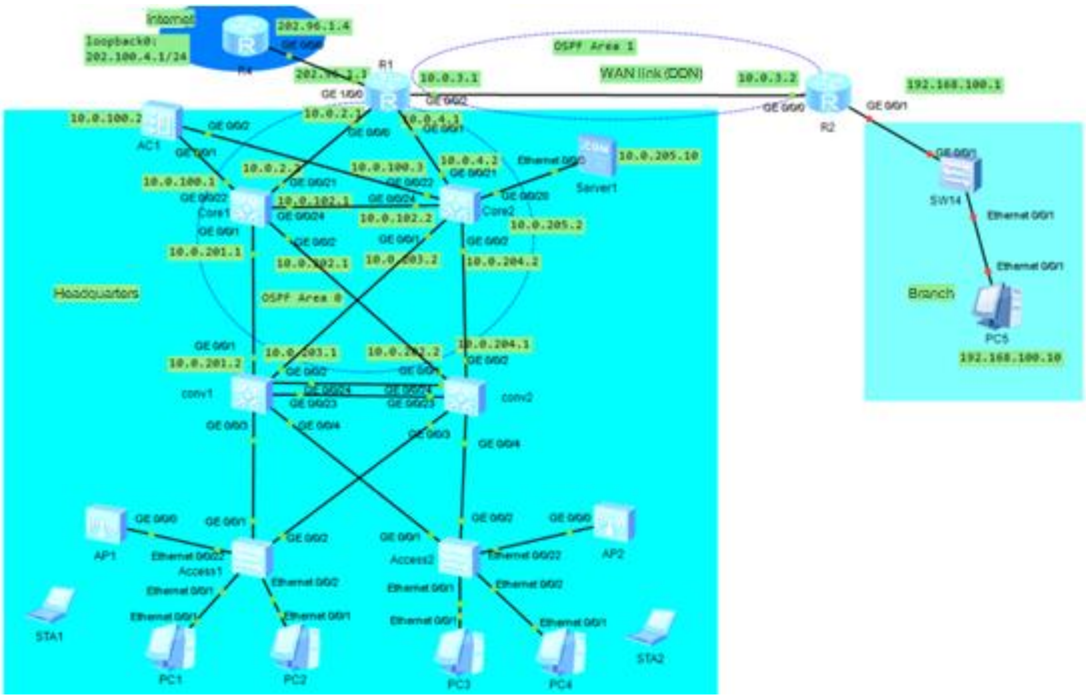

Figure 1 Typical campus network topology

Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
```

```
[Huawei]sysname R1
[R1]int lo0
[R1-LoopBack0] ip address 10.0.1.1 32
[R1-LoopBack0] quit
[R1]
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0] ip address 10.0.2.1 24
[R1-GigabitEthernet0/0/0] quit
[R1]
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1] ip address 10.0.4.1 24
[R1-GigabitEthernet0/0/1] quit
[R1]
[R1]int g0/0/2
[R1-GigabitEthernet0/0/2] ip address 10.0.3.1 24
[R1-GigabitEthernet0/0/2] quit
[R1]
[R1]int g1/0/0
[R1-GigabitEthernet1/0/0] ip address 202.96.1.1 24
[R1-GigabitEthernet1/0/0] quit
[R1]
[R1]ip route-static 0.0.0.0 0.0.0.0 202.96.1.4
[R1]
```

- Initial configuration of R4

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R4
[R4]int lo0
[R4-LoopBack0] ip address 202.100.4.1 24
[R4-LoopBack0] quit
[R4]
[R4]int g0/0/0
[R4-GigabitEthernet0/0/0]
[R4-GigabitEthernet0/0/0] ip address 202.96.1.4 24
[R4-GigabitEthernet0/0/0] quit
[R4]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int lo0
[R2-LoopBack0]ip address 10.0.20.1 32
[R2-LoopBack0]quit
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.0.3.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 192.168.100.1 24
[R2-GigabitEthernet0/0/1]quit
[R2]
```

- Initial configuration of AC1

```
<AC6005>system-view
Enter system view, return user view with Ctrl+Z.
```

```
[AC6005]st
[AC6005]sys
[AC6005]sysname AC1
[AC1]vlan 100
Info: This operation may take a few seconds. Please wait for a moment...done.
[AC1-vlan100]quit
[AC1]int lo0
[AC1-LoopBack0] ip address 10.10.10.10. 24
[AC1-LoopBack0]quit
[AC1]int vlan 100
[AC1-Vlanif100]ip address 10.10.100.2 24
[AC1-Vlanif100]quit
[AC1]int g0/0/1
[AC1-GigabitEthernet0/0/1]port link-type access
[AC1-GigabitEthernet0/0/1]port default vlan 100
[AC1-GigabitEthernet0/0/1]quit
[AC1]int g0/0/2
[AC1-GigabitEthernet0/0/2]port link-type access
[AC1-GigabitEthernet0/0/2]port default vlan 100
[AC1-GigabitEthernet0/0/2]quit
[AC1]
[AC1]ip route-static 0.0.0.0 0.0.0.0 10.0.100.1
[AC1]
```

- Initial configuration of Core1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname core1
[core1]
[core1]vlan batch 100 101 102 201 202
Info: This operation may take a few seconds. Please wait for a moment...done.
[core1]
[core1]undo stp enable
Warning: The global STP state will be changed. Continue? [Y/N]y
Info: This operation may take a few seconds. Please wait for a moment...done.
[core1]
[core1]int lo0
[core1-LoopBack0] ip address 10.0.10.1 32
[core1-LoopBack0] quit
[core1]int  g0/0/21
[core1-GigabitEthernet0/0/21] port link-type access
[core1-GigabitEthernet0/0/21] port default vlan 101
[core1-GigabitEthernet0/0/21] quit
[core1]int  g0/0/22
[core1-GigabitEthernet0/0/22] port link-type access
[core1-GigabitEthernet0/0/22] port default vlan 100
[core1-GigabitEthernet0/0/22] quit
[core1]int  g0/0/24
[core1-GigabitEthernet0/0/24] port link-type access
[core1-GigabitEthernet0/0/24] port default vlan 102
[core1-GigabitEthernet0/0/24] quit
[core1]int  g0/0/1
[core1-GigabitEthernet0/0/1] port link-type access
[core1-GigabitEthernet0/0/1] port default vlan 201
[core1-GigabitEthernet0/0/1] quit
```

```
[core1]int   g0/0/2
[core1-GigabitEthernet0/0/2] port link-type access
[core1-GigabitEthernet0/0/2] port default vlan 202
[core1-GigabitEthernet0/0/2] quit
[core1]
[core1]int vlanif 100
[core1-Vlanif100] ip address 10.0.100.1 24
[core1-Vlanif100] quit
[core1]
[core1]int vlanif 101
[core1-Vlanif101]ip address 10.0.2.2 24
[core1-Vlanif101] quit
[core1]
[core1]int vlanif 102
[core1-Vlanif102] ip address 10.0.102.1 24
[core1-Vlanif102] quit
[core1]
[core1]int vlanif 201
[core1-Vlanif201]ip address 10.0.201.1 24
[core1-Vlanif201] quit
[core1]
[core1]int vlanif 202
[core1-Vlanif202] ip address 10.0.202.1 24
[core1-Vlanif202] quit
[core1]
[core1]ip route-static 10.10.10.0 255.255.255.0 10.0.100.2
```

- Initial configuration of Core2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname core2
[core2]
[core2]vlan batch 100 102 104 203 204 205
Info: This operation may take a few seconds. Please wait for a moment...done.
[core2]
[core2]undo stp enable
Warning: The global STP state will be changed. Continue? [Y/N]y
Info: This operation may take a few seconds. Please wait for a moment...done.
[core2]
[core2]int lo0
[core2-LoopBack0] ip address 10.0.10.2 32
[core2-LoopBack0] quit
[core2]int   g0/0/22
[core2-GigabitEthernet0/0/22] port link-type access
[core2-GigabitEthernet0/0/22] port default vlan 100
[core2-GigabitEthernet0/0/22] quit
[core2]
[core2]int   g0/0/21
[core2-GigabitEthernet0/0/21] port link-type access
[core2-GigabitEthernet0/0/21] port default vlan 104
[core2-GigabitEthernet0/0/21] quit
[core2]
[core2]int   g0/0/24
[core2-GigabitEthernet0/0/24] port link-type access
[core2-GigabitEthernet0/0/24] port default vlan 102
[core2-GigabitEthernet0/0/24] quit
```

```
[core2]
[core2]int   g0/0/1
[core2-GigabitEthernet0/0/1] port link-type access
[core2-GigabitEthernet0/0/1] port default vlan 203
[core2-GigabitEthernet0/0/1] quit
[core2]
[core2]int   g0/0/2
[core2-GigabitEthernet0/0/2] port link-type access
[core2-GigabitEthernet0/0/2] port default vlan 204
[core2-GigabitEthernet0/0/2] quit
[core2]
[core2]int   g0/0/20
[core2-GigabitEthernet0/0/20] port link-type access
[core2-GigabitEthernet0/0/20] port default vlan 205
[core2-GigabitEthernet0/0/20] quit
[core2]
[core2]int vlanif 102
[core2-Vlanif102] ip address 10.0.102.2 24
[core2-Vlanif102] quit
[core2]
[core2]int vlanif 104
[core2-Vlanif104] ip address 10.0.4.2 24
[core2-Vlanif104] quit
[core2]
[core2]int vlanif 100
[core2-Vlanif100] ip address 10.0.100.3 24
[core2-Vlanif100] quit
[core2]
[core2]int vlanif 203
[core2-Vlanif203] ip address 10.0.203.2 24
[core2-Vlanif203] quit
[core2]
[core2]int vlanif 205
[core2-Vlanif205] ip address 10.0.205.2 24
[core2-Vlanif205] quit
[core2]
[core2]
[core2]int vlanif 204
[core2-Vlanif204] ip address 10.0.204.2 24
[core2-Vlanif204] quit
```

- Initial configuration of Conv1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname Conv1
[Conv1]
[Conv1]vlan batch 201 203
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv1]
[Conv1]int lo0
[Conv1-LoopBack0] ip address 10.0.10.3 32
[Conv1-LoopBack0] quit
[Conv1]
```

```
[Conv1]int  g0/0/1
[Conv1-GigabitEthernet0/0/1] port link-type access
[Conv1-GigabitEthernet0/0/1] port default vlan 201
[Conv1-GigabitEthernet0/0/1] undo stp enable
[Conv1-GigabitEthernet0/0/1] quit
[Conv1]
[Conv1]int  g0/0/2
[Conv1-GigabitEthernet0/0/2] port link-type access
[Conv1-GigabitEthernet0/0/2] port default vlan 203
[Conv1-GigabitEthernet0/0/2] undo stp enable
[Conv1-GigabitEthernet0/0/2] quit
[Conv1]
[Conv1]int vlanif 201
[Conv1-Vlanif201]  ip address 10.0.201.2 24
[Conv1-Vlanif201] quit
[Conv1]
[Conv1]int vlanif 203
[Conv1-Vlanif203] ip address 10.0.203.1 24
[Conv1-Vlanif203] quit
[Conv1]
```

- Initial configuration of Conv2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname Conv2
[Conv2]
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname Conv2
[Conv2]
[Conv2]vlan batch 202 204
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv2]
[Conv2]int lo0
[Conv2-LoopBack0] ip address 10.0.10.4 32
[Conv2-LoopBack0] quit
[Conv2]
[Conv2]int  g0/0/1
[Conv2-GigabitEthernet0/0/1] port link-type access
[Conv2-GigabitEthernet0/0/1] port default vlan 202
[Conv2-GigabitEthernet0/0/1]undo stp enable
[Conv2-GigabitEthernet0/0/1] quit
[Conv2]
[Conv2]int  g0/0/2
[Conv2-GigabitEthernet0/0/2] port link-type access
[Conv2-GigabitEthernet0/0/2] port default vlan 204
[Conv2-GigabitEthernet0/0/2]undo stp enable
[Conv2-GigabitEthernet0/0/2] quit
[Conv2]
[Conv2]int vlanif 202
[Conv2-Vlanif202]  ip address 10.0.202.2 24
[Conv2-Vlanif202] quit
[Conv2]
[Conv2]int vlanif 204
```

```
[Conv2-Vlanif204]   ip address 10.0.204.1 24
[Conv2-Vlanif204]   quit
[Conv2]
```

- Initial configuration of Access1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname Access1
[Access1]
```

- Initial configuration of Access2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname Access2
[Access2]
```

Task Implementation

1 Checking Basic Configurations

Test the network connectivity.

- Test the connectivity of all directly connected interfaces on Core1.

```
Ping -c 2 10.0.2.1
Ping -c 2 10.0.102.2
Ping -c 2 10.0.100.1
Ping -c 2 10.0.201.2
Ping -c 2 10.0.202.2

<core1>Ping -c 2 10.0.2.1
  PING 10.0.2.1: 56   data bytes, press CTRL_C to break
    Reply from 10.0.2.1: bytes=56 Sequence=1 ttl=255 time=50 ms
    Reply from 10.0.2.1: bytes=56 Sequence=2 ttl=255 time=40 ms

<core1>Ping -c 2 10.0.102.2
  PING 10.0.102.2: 56   data bytes, press CTRL_C to break
    Reply from 10.0.102.2: bytes=56 Sequence=1 ttl=255 time=60 ms
    Reply from 10.0.102.2: bytes=56 Sequence=2 ttl=255 time=50 ms

<core1>Ping -c 2 10.0.100.1
  PING 10.0.100.1: 56   data bytes, press CTRL_C to break
    Reply from 10.0.100.1: bytes=56 Sequence=1 ttl=255 time=1 ms
    Reply from 10.0.100.1: bytes=56 Sequence=2 ttl=255 time=1 ms

<core1>Ping -c 2 10.0.201.2
  PING 10.0.201.2: 56   data bytes, press CTRL_C to break
    Reply from 10.0.201.2: bytes=56 Sequence=1 ttl=255 time=60 ms
    Reply from 10.0.201.2: bytes=56 Sequence=2 ttl=255 time=50 ms

<core1>Ping -c 2 10.0.202.2
  PING 10.0.202.2: 56   data bytes, press CTRL_C to break
    Reply from 10.0.202.2: bytes=56 Sequence=1 ttl=255 time=50 ms
    Reply from 10.0.202.2: bytes=56 Sequence=2 ttl=255 time=30 ms

<core1>
```

- Test the connectivity of all directly connected interfaces on Core2.

```
Ping -c 2 10.0.100.2
Ping -c 2 10.0.4.1
Ping -c 2 10.0.205.10
Ping -c 2 10.0.102.1
Ping -c 2 10.0.203.1
Ping -c 2 10.0.204.1

[core2]Ping -c 2 10.0.100.2
  PING 10.0.100.2: 56  data bytes, press CTRL_C to break
    Reply from 10.0.100.2: bytes=56 Sequence=1 ttl=255 time=60 ms
    Reply from 10.0.100.2: bytes=56 Sequence=2 ttl=255 time=50 ms

[core2]Ping -c 2 10.0.4.1
  PING 10.0.4.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.4.1: bytes=56 Sequence=1 ttl=255 time=60 ms
    Reply from 10.0.4.1: bytes=56 Sequence=2 ttl=255 time=50 ms


[core2]Ping -c 2 10.0.205.10
  PING 10.0.205.10: 56  data bytes, press CTRL_C to break
    Reply from 10.0.205.10: bytes=56 Sequence=1 ttl=255 time=30 ms
    Reply from 10.0.205.10: bytes=56 Sequence=2 ttl=255 time=10 ms

[core2]Ping -c 2 10.0.102.1
  PING 10.0.102.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.102.1: bytes=56 Sequence=1 ttl=255 time=60 ms
    Reply from 10.0.102.1: bytes=56 Sequence=2 ttl=255 time=20 ms

[core2]Ping -c 2 10.0.203.1
  PING 10.0.203.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.203.1: bytes=56 Sequence=1 ttl=255 time=50 ms
    Reply from 10.0.203.1: bytes=56 Sequence=2 ttl=255 time=40 ms

[core2]Ping -c 2 10.0.204.1
  PING 10.0.204.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.204.1: bytes=56 Sequence=1 ttl=255 time=60 ms
    Reply from 10.0.204.1: bytes=56 Sequence=2 ttl=255 time=30 ms

[core2]
```

- Test the connectivity of all directly connected interfaces on R1.

```
<R1>ping -c 2 202.96.1.4
  PING 202.96.1.4: 56  data bytes, press CTRL_C to break
    Reply from 202.96.1.4: bytes=56 Sequence=1 ttl=255 time=90 ms
    Reply from 202.96.1.4: bytes=56 Sequence=2 ttl=255 time=20 ms

<R1>ping -c 2 10.0.3.2
  PING 10.0.3.2: 56  data bytes, press CTRL_C to break
    Reply from 10.0.3.2: bytes=56 Sequence=1 ttl=255 time=60 ms
    Reply from 10.0.3.2: bytes=56 Sequence=2 ttl=255 time=10 ms

<R1>
```

2 Configuring OSPF

The headquarters is in OSPF area 0, and the branch is in OSPF area 1.

- OSPF configuration on R1

```
[R1]ospf 1 router-id 10.0.1.1
[core1-ospf-1]import-route static
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]network 10.0.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]network 10.0.3.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]network 10.0.4.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]quit
[R1-ospf-1]
[R1-ospf-1]area 1
[R1-ospf-1-area-0.0.0.1]
[R1-ospf-1-area-0.0.0.1]network 10.0.2.1 0.0.0.0
[R1-ospf-1-area-0.0.0.1]quit
[R1-ospf-1]quit
[R1]
```

- OSPF configuration on R2

```
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ospf network-type p2p
[R2-GigabitEthernet0/0/0]quit
[R2]
[R2]ospf 1 router-id 10.0.20.1
[R2-ospf-1]
[R2-ospf-1]area 1
[R2-ospf-1-area-0.0.0.1]network 10.0.20.1 0.0.0.0
[R2-ospf-1-area-0.0.0.1]network 10.0.3.2 0.0.0.0
[R2-ospf-1-area-0.0.0.1]network 192.168.100.1 0.0.0.0
[R2-ospf-1-area-0.0.0.1]quit
[R2-ospf-1]quit
[R2]
```

- OSPF configuration on Core1

```
[core1]ospf 1 router-id 10.0.10.1
[core1-ospf-1]area 0
[core1-ospf-1-area-0.0.0.0]network 10.0.10.1 0.0.0.0
[core1-ospf-1-area-0.0.0.0]network 10.0.100.1 0.0.0.0
[core1-ospf-1-area-0.0.0.0]network 10.0.2.2 0.0.0.0
[core1-ospf-1-area-0.0.0.0]network 10.0.102.1 0.0.0.0
[core1-ospf-1-area-0.0.0.0]network 10.0.201.1 0.0.0.0
[core1-ospf-1-area-0.0.0.0]network 10.0.202.1 0.0.0.0
[core1-ospf-1-area-0.0.0.0]quit
[core1-ospf-1]quit
[core1]
```

- OSPF configuration on Core2

```
[core2]ospf 1 router-id 10.0.10.2
[core2-ospf-1]area 0
[core2-ospf-1-area-0.0.0.0]network 10.0.10.2 0.0.0.0
[core2-ospf-1-area-0.0.0.0]network 10.0.100.3 0.0.0.0
[core2-ospf-1-area-0.0.0.0]network 10.0.102.2 0.0.0.0
[core2-ospf-1-area-0.0.0.0]network 10.0.4.2 0.0.0.0
[core2-ospf-1-area-0.0.0.0]network 10.0.203.2 0.0.0.0
```

```
[core2-ospf-1-area-0.0.0.0]network 10.0.204.2 0.0.0.0
[core2-ospf-1-area-0.0.0.0]network 10.0.205.2 0.0.0.0
[core2-ospf-1-area-0.0.0.0]quit
[core2-ospf-1]
[core2]
```

- OSPF configuration on Conv1

```
[Conv1]
[Conv1]ospf 1 router-id 10.0.10.3
[Conv1-ospf-1]area 0
[Conv1-ospf-1-area-0.0.0.0]network 10.0.10.3 0.0.0.0
[Conv1-ospf-1-area-0.0.0.0]network   10.0.201.2 0.0.0.0
[Conv1-ospf-1-area-0.0.0.0]network 10.0.203.1 0.0.0.0
[Conv1-ospf-1-area-0.0.0.0]
[Conv1-ospf-1-area-0.0.0.0]quit
[Conv1-ospf-1]quit
[Conv1]
```

- OSPF configuration on Conv2

```
[Conv2]ospf 1 router-id 10.0.10.4
[Conv2-ospf-1]area 0
[Conv2-ospf-1-area-0.0.0.0]network 10.0.10.4 0.0.0.0
[Conv2-ospf-1-area-0.0.0.0]network   10.0.202.2 0.0.0.0
[Conv2-ospf-1-area-0.0.0.0]network 10.0.204.1 0.0.0.0
[Conv2-ospf-1-area-0.0.0.0]quit
[Conv2-ospf-1]quit
[Conv2]
```

- Check the OSPF neighbor relationships on Core1.

```
[core1]display ospf peer brief

  OSPF Process 1 with Router ID 10.0.10.1
    Peer Statistic Information
 ----------------------------------------------------------------------------
 Area Id          Interface                       Neighbor id      State
 0.0.0.0          Vlanif100                       10.0.10.2        Full
 0.0.0.0          Vlanif101                       10.0.1.1         Full
 0.0.0.0          Vlanif102                       10.0.10.2        Full
 0.0.0.0          Vlanif201                       10.0.10.3        Full
 0.0.0.0          Vlanif202                       10.0.10.4        Full
 ----------------------------------------------------------------------------
[core1]
```

- Check the OSPF neighbor relationships on Core2.

```
[core2]display ospf peer brief

  OSPF Process 1 with Router ID 10.0.10.2
    Peer Statistic Information
 ----------------------------------------------------------------------------
 Area Id          Interface                       Neighbor id      State
 0.0.0.0          Vlanif102                       10.0.10.1        Full
 0.0.0.0          Vlanif104                       10.0.1.1         Full
 0.0.0.0          Vlanif100                       10.0.10.1        Full
 0.0.0.0          Vlanif203                       10.0.10.3        Full
```

```
0.0.0.0              Vlanif204                               10.0.10.4           Full
----------------------------------------------------------------------------
[core2]
```

- Check the OSPF neighbor relationships on R1.

```
<R1>display ospf peer brief

  OSPF Process 1 with Router ID 10.0.1.1
     Peer Statistic Information
----------------------------------------------------------------------------
Area Id              Interface                              Neighbor id          State
0.0.0.0              GigabitEthernet0/0/0                   10.0.10.1            Full
0.0.0.0              GigabitEthernet0/0/1                   10.0.10.2            Full
0.0.0.1              GigabitEthernet0/0/2                   10.0.20.1            Full
----------------------------------------------------------------------------
<R1>
```

- Check the OSPF neighbor relationships on R2.

```
<R2>display ospf peer brief

  OSPF Process 1 with Router ID 10.0.20.1
     Peer Statistic Information
----------------------------------------------------------------------------
Area Id              Interface                              Neighbor id          State
0.0.0.1              GigabitEthernet0/0/0                   10.0.1.1             Full
----------------------------------------------------------------------------
<R2>
```

3 Configuring MSTP

Create VLANs 10 and 20 on aggregation switches Conv1 and Conv2 as well as access switches Access1 and Access2. Configure an MSTP region named **hcip**, and create two instances **Instance 1** and **Instance 2**. Map VLAN 10 to **Instance 1** and VLAN 20 to **Instance 2**. Plan Conv1 as the primary root bridge of MSTI 1 and secondary root bridge of MSTI 2; plan Conv2 as the primary root bridge of MSTI 2 and the secondary root bridge of MSTI 1.

1. Create VLANs.

```
[Conv1]vlan batch 10 20
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv1]

[Conv2]vlan batch 10 20
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv2]

[Access2]vlan batch 10 20
Info: This operation may take a few seconds. Please wait for a moment...done.
[Access2]

[Access1]vlan batch 10 20
Info: This operation may take a few seconds. Please wait for a moment...done.
[Access1]
```

2. Configure an Eth-Trunk between aggregation switches Conv1 and Conv2, and allow packets from the corresponding VLANs to pass through.

```
[Conv1]
```

```
[Conv1]int Eth-Trunk 1
[Conv1-Eth-Trunk1]port link-type trunk
[Conv1-Eth-Trunk1]port trunk allow-pass vlan 10 20
[Conv1-Eth-Trunk1]quit
[Conv1]
[Conv1]int g0/0/23
[Conv1-GigabitEthernet0/0/23]eth-trunk 1
[Conv1-GigabitEthernet0/0/23]quit
[Conv1]
[Conv1]int g0/0/24
[Conv1-GigabitEthernet0/0/24]eth-trunk 1
[Conv1-GigabitEthernet0/0/24]quit
[Conv1]


[Conv2]
[Conv2]int Eth-Trunk 1
[Conv2-Eth-Trunk1]port link-type trunk
[Conv2-Eth-Trunk1]port trunk allow-pass vlan 10 20
[Conv2-Eth-Trunk1]quit
[Conv2]
[Conv2]int g0/0/23
[Conv2-GigabitEthernet0/0/23]eth-trunk 1
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv2-GigabitEthernet0/0/23]quit
[Conv2]
[Conv2]int g0/0/24
[Conv2-GigabitEthernet0/0/24]eth-trunk 1
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv2-GigabitEthernet0/0/24]quit
[Conv2]


[Conv1]display eth-trunk 1
Eth-Trunk1's state information is:
WorkingMode: NORMAL             Hash arithmetic: According to SIP-XOR-DIP
Least Active-linknumber: 1    Max Bandwidth-affected-linknumber: 8
Operate status: up              Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
PortName                          Status        Weight
GigabitEthernet0/0/23             Up            1
GigabitEthernet0/0/24             Up            1

[Conv1]

[Conv2]display eth-trunk 1
Eth-Trunk1's state information is:
WorkingMode: NORMAL             Hash arithmetic: According to SIP-XOR-DIP
Least Active-linknumber: 1    Max Bandwidth-affected-linknumber: 8
Operate status: up              Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
PortName                          Status        Weight
GigabitEthernet0/0/23             Up            1
GigabitEthernet0/0/24             Up            1
```

```
[Conv2]
```

3. Configure the other interconnection interfaces as trunk interfaces and allow packets from the corresponding VLANs to pass through.
   - Configure Conv1.

```
[Conv1]int g0/0/3
[Conv1-GigabitEthernet0/0/3]port link-type trunk
[Conv1-GigabitEthernet0/0/3]port trunk allow-pass vlan 10 20
[Conv1-GigabitEthernet0/0/3]quit
[Conv1]int g0/0/4
[Conv1-GigabitEthernet0/0/4]port link-type trunk
[Conv1-GigabitEthernet0/0/4]port trunk allow-pass vlan 10 20
[Conv1-GigabitEthernet0/0/4]quit
[Conv1]
```

   - Configure Conv2.

```
[Conv2]int g0/0/3
[Conv2-GigabitEthernet0/0/3]port link-type trunk
[Conv2-GigabitEthernet0/0/3]port trunk allow-pass vlan 10 20
[Conv2-GigabitEthernet0/0/3]quit
[Conv2]int g0/0/4
[Conv2-GigabitEthernet0/0/4]port link-type trunk
[Conv2-GigabitEthernet0/0/4]port trunk allow-pass vlan 10 20
[Conv2-GigabitEthernet0/0/4]quit
[Conv2]
```

   - Configure Access1.

```
[Access1]int g0/0/1
[Access1-GigabitEthernet0/0/1]port link-type trunk
[Access1-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 20
[Access1-GigabitEthernet0/0/1]quit
[Access1]int g0/0/2
[Access1-GigabitEthernet0/0/2]port link-type trunk
[Access1-GigabitEthernet0/0/2]port trunk allow-pass vlan 10 20
[Access1-GigabitEthernet0/0/2]quit
[Access1]
```

   - Configure Access2.

```
[Access2]int g0/0/1
[Access2-GigabitEthernet0/0/1]port link-type trunk
[Access2-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 20
[Access2-GigabitEthernet0/0/1]quit
[Access2]int g0/0/2
[Access2-GigabitEthernet0/0/2]port link-type trunk
[Access2-GigabitEthernet0/0/2]port trunk allow-pass vlan 10 20
[Access2-GigabitEthernet0/0/2]quit
[Access2]
```

4. Change the working mode from STP to MSTP.

```
[Conv1]stp mode mstp

[Conv2]stp mode mstp

[Access1]stp mode mstp
```

```
[Access2]stp mode mstp
```

5. Configure MSTP.

```
[Conv1]stp region-configuration
[Conv1-mst-region]region-name hcip
[Conv1-mst-region]revision-level 1
[Conv1-mst-region] instance 1 vlan 10
[Conv1-mst-region]instance 2 vlan 20
[Conv1-mst-region]active region-configuration
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv1-mst-region]quit
[Conv1]

[Conv2]stp region-configuration
[Conv2-mst-region]region-name hcip
[Conv2-mst-region]revision-level 1
[Conv2-mst-region] instance 1 vlan 10
[Conv2-mst-region]instance 2 vlan 20
[Conv2-mst-region]active region-configuration
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv2-mst-region]quit
[Conv2]

[Access1]stp region-configuration
[Access1-mst-region]region-name hcip
[Access1-mst-region]revision-level 1
[Access1-mst-region] instance 1 vlan 10
[Access1-mst-region]instance 2 vlan 20
[Access1-mst-region]active region-configuration
Info: This operation may take a few seconds. Please wait for a moment...done.
[Access1-mst-region]quit
[Access1]

[Access2]stp region-configuration
[Access2-mst-region]region-name hcip
[Access2-mst-region]revision-level 1
[Access2-mst-region] instance 1 vlan 10
[Access2-mst-region]instance 2 vlan 20
[Access2-mst-region]active region-configuration
Info: This operation may take a few seconds. Please wait for a moment...done.
[Access2-mst-region]quit
[Access2]
```

6. Check the mappings between MSTI instances and VLANs on Conv1.

```
 [Conv1]display stp region-configuration
 Oper configuration
   Format selector      :0
   Region name          :hcip
   Revision level       :1

   Instance    VLANs Mapped
      0         1 to 9, 11 to 19, 21 to 4094
      1         10
      2         20
[Conv1]
```

7. Configure Conv1 as the primary root bridge of MSTI 1 and the secondary root bridge of MSTI 2.

```
[Conv1]stp instance 1 root primary
[Conv1]stp instance 2 root secondary
```

8. Configure Conv2 as the primary root bridge of MSTI 2 and the secondary root bridge of MSTI 1.

```
[Conv2]stp instance 1 root secondary
[Conv2]stp instance 2 root primary
```

9. Check the status and statistics of MSTI 1 on Conv1.

```
[Conv1]display stp instance 1 brief
 MSTID   Port                      Role  STP State     Protection
    1    GigabitEthernet0/0/3      DESI  FORWARDING      NONE
    1    GigabitEthernet0/0/4      DESI  FORWARDING      NONE
    1    Eth-Trunk1                DESI  FORWARDING      NONE
[Conv1]
```

The command output shows that all interfaces on Conv1 are designated interfaces, and Conv1 is the root bridge of MSTI 1.

10. Check the status and statistics of MSTI 2 on Conv2.

```
[Conv2]display stp instance 2 brief
 MSTID   Port                      Role  STP State     Protection
    2    GigabitEthernet0/0/3      DESI  FORWARDING      NONE
    2    GigabitEthernet0/0/4      DESI  FORWARDING      NONE
    2    Eth-Trunk1                DESI  FORWARDING      NONE
[Conv2]
```

The command output shows that all interfaces on Conv2 are designated interfaces, and Conv2 is the root bridge of MSTI 2.

4 Configuring VRRP

Create VLANIF 10 and VLANIF 20 on both Conv1 and Conv2, and add VLANIF 10 to VRRP group 10 and VLANIF 20 to VRRP group 20. Configure VRRP priorities so that Conv1 in VLAN 10 and Conv2 in VLAN 20 both function as the VRRP master.

1. Create VLANIF interfaces.

```
[Conv1]interface Vlanif10
[Conv1-Vlanif10]ip address 172.16.10.1 255.255.255.0
[Conv1-Vlanif10]
[Conv1]
[Conv1]int Vlanif20
[Conv1-Vlanif20]ip address 172.16.20.1 255.255.255.0
[Conv1-Vlanif20]quit
[Conv1]

[Conv2]interface Vlanif10
[Conv2-Vlanif10]ip address 172.16.10.2 255.255.255.0
[Conv2-Vlanif10]quit
[Conv2]
[Conv2]interface Vlanif20
[Conv2-Vlanif20]ip address 172.16.20.2 255.255.255.0
[Conv2-Vlanif20]quit
[Conv2]
```

2. Add the VLANIF interface to OSPF area 0.

```
[Conv1]ospf 1
[Conv1-ospf-1]area 0
[Conv1-ospf-1-area-0.0.0.0]network 172.16.10.1 0.0.0.0
[Conv1-ospf-1-area-0.0.0.0]network 172.16.20.1 0.0.0.0
[Conv1-ospf-1-area-0.0.0.0]quit
[Conv1-ospf-1]quit
[Conv1]

[Conv2]ospf 1
[Conv2-ospf-1]area 0
[Conv2-ospf-1-area-0.0.0.0]network 172.16.10.2 0.0.0.0
[Conv2-ospf-1-area-0.0.0.0]network 172.16.20.2 0.0.0.0
[Conv2-ospf-1-area-0.0.0.0]quit
[Conv2-ospf-1]quit
[Conv2]
```

3. Configure VRRP groups on Conv1.

```
[Conv1]interface Vlanif 10
[Conv1-Vlanif10]vrrp vrid 10 virtual-ip 172.16.10.254
[Conv1-Vlanif10]vrrp vrid 10 priority 120
[Conv1-Vlanif10]quit
[Conv1]
[Conv1]interface Vlanif 20
[Conv1-Vlanif20]vrrp vrid 20 virtual-ip 172.16.20.254
[Conv1-Vlanif20]quit
[Conv1]
```

Set the VRRP priority to 120 for Conv1 in VLAN 10, and use the default value 100 for Conv1 in VLAN 20.

4. Configure VRRP groups on Conv2.

```
[Conv2]interface Vlanif10
[Conv2-Vlanif10]vrrp vrid 10 virtual-ip 172.16.10.254
[Conv2-Vlanif10]quit
[Conv2]int Vlanif20
[Conv2-Vlanif20]vrrp vrid 20 virtual-ip 172.16.20.254
[Conv2-Vlanif20]vrrp vrid 20 priority 120
[Conv2-Vlanif20]quit
```

Set the VRRP priority to 120 for Conv2 in VLAN 20, and use the default value 100 for Conv2 in VLAN 10.

5. Check the VRRP status.

```
[Conv1]display vrrp brief
VRID   State          Interface                 Type      Virtual IP
----------------------------------------------------------------
10     Master         Vlanif10                  Normal    172.16.10.254
20     Backup         Vlanif20                  Normal    172.16.20.254
----------------------------------------------------------------
Total:2      Master:1      Backup:1      Non-active:0
[Conv1]

[Conv2]display vrrp brief
VRID   State          Interface                 Type      Virtual IP
----------------------------------------------------------------
10     Backup         Vlanif10                  Normal    172.16.10.254
```

```
20      Master          Vlanif20                     Normal     172.16.20.254
----------------------------------------------------------------
Total:2         Master:1        Backup:1        Non-active:0
[Conv2]
```

The VRRP status is the same as expected.

5 Configuring DHCP

Configure the PCs to obtain IP address through DHCP. In the topology, PC1 and PC3 belong to VLAN 10, and PC2 and PC4 belong to VLAN 20.

1. Configure Access1.

```
[Access1]int e0/0/1
[Access1-Ethernet0/0/1]port link-type access
[Access1-Ethernet0/0/1]port default vlan 10
[Access1-Ethernet0/0/1] quit
[Access1]int e0/0/2
[Access1-Ethernet0/0/2]port link-type access
[Access1-Ethernet0/0/2]port default vlan 20
[Access1-Ethernet0/0/2] quit
[Access1]
```

2. Configure Access2.

```
[Access2]int e0/0/1
[Access2-Ethernet0/0/1]port link-type access
[Access2-Ethernet0/0/1]port default vlan 10
[Access2-Ethernet0/0/1]quit
[Access2]
[Access2]int e0/0/2
[Access2-Ethernet0/0/2]port link-type access
[Access2-Ethernet0/0/2]port default vlan 20
[Access2-Ethernet0/0/2]quit
[Access2]
```

3. Configure DHCP on Conv1 and use the interface address pool.

```
[Conv1]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.

[Conv1]int Vlanif 10
[Conv1-Vlanif10]dhcp select interface
[Conv1-Vlanif10]quit
[Conv1]
[Conv1]int Vlanif 20
[Conv1-Vlanif20]dhcp select interface
[Conv1-Vlanif20]quit
[Conv1]
```

4. Configure DHCP on Conv2 and use the interface address pool.

```
[Conv2]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.

[Conv2]int Vlanif 20
[Conv2-Vlanif20]dhcp select interface
[Conv2-Vlanif20]quit
[Conv2]
```

```
[Conv2]int Vlanif 10
[Conv2-Vlanif10]dhcp select interface
[Conv2-Vlanif10]quit
[Conv2]
```

5. Check the addresses of PCs. PC1 and PC4 are used as examples.
   - PC1

```
PC>ipconfig

Link local IPv6 address...........: fe80::5689:98ff:fedd:491c
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 172.16.10.253
Subnet mask.......................: 255.255.255.0
Gateway...........................: 172.16.10.1
Physical address..................: 54-89-98-DD-49-1C
DNS server........................:

PC>ping 10.0.10.1 -c 2

Ping 10.0.10.1: 32 data bytes, Press Ctrl_C to break
From 10.0.10.1: bytes=32 seq=1 ttl=254 time=188 ms
From 10.0.10.1: bytes=32 seq=2 ttl=254 time=93 ms
```

   - PC4

```
PC>ipconfig

Link local IPv6 address...........: fe80::5689:98ff:fe89:4f21
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 172.16.20.252
Subnet mask.......................: 255.255.255.0
Gateway...........................: 172.16.20.2
Physical address..................: 54-89-98-89-4F-21
DNS server........................:

PC>ping 10.0.10.1 -c 2

Ping 10.0.10.1: 32 data bytes, Press Ctrl_C to break
From 10.0.10.1: bytes=32 seq=1 ttl=254 time=110 ms
From 10.0.10.1: bytes=32 seq=2 ttl=254 time=62 ms
```

6 Configuring AP Onboarding

Configure Conv1 as a DHCP server to ensure that users and APs can obtain IP addresses. Add three VLANs: VLAN 60, VLAN 80, and VLAN 90. VLAN 60 provides an IP address pool for APs; VLAN 80 provides an IP address pool for wireless connections of enterprise employees (SSID: Employee); VLAN 90 provides an IP address pool for wireless connections of guests (SSID: Guest).

1. Create VLAN 60, VLAN 80, and VLAN 90 on Conv1, Conv2, Access1, and Access2, and permit packets from the three VLANs on all trunk interfaces.
   - Conv1

```
[Conv1]vlan batch 60 80 90
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv1]

[Conv1]int g0/0/3
[Conv1-GigabitEthernet0/0/3]port trunk allow-pass vlan 10 20 60 80 90
[Conv1-GigabitEthernet0/0/3]quit
[Conv1]

[Conv1]int g0/0/4
[Conv1-GigabitEthernet0/0/4]port trunk allow-pass vlan 10 20 60 80 90
[Conv1-GigabitEthernet0/0/4]quit
[Conv1]

[Conv1]int Eth-Trunk 1
[Conv1-Eth-Trunk1]port trunk allow-pass vlan 10 20 60 80 90
[Conv1-Eth-Trunk1]quit
[Conv1]

[Conv1]interface Vlanif60
[Conv1-Vlanif60]ip address 172.16.60.1 255.255.255.0
[Conv1-Vlanif60] quit
[Conv1]
[Conv1]interface Vlanif80
[Conv1-Vlanif80] ip address 172.16.80.1 255.255.255.0
[Conv1-Vlanif80] quit
[Conv1]
[Conv1]interface Vlanif90
[Conv1-Vlanif90] ip address 172.16.90.1 255.255.255.0
[Conv1-Vlanif90] quit
[Conv1]
# Add VLANIF 60, VLANIF 80, and VLANIF 90 to OSPF area 0.
[Conv1]ospf 1
[Conv1-ospf-1]area 0
[Conv1-ospf-1-area-0.0.0.0] network 172.16.60.1 0.0.0.0
[Conv1-ospf-1-area-0.0.0.0] network 172.16.80.1 0.0.0.0
[Conv1-ospf-1-area-0.0.0.0] network 172.16.90.1 0.0.0.0
[Conv1-ospf-1-area-0.0.0.0]quit
[Conv1-ospf-1]quit
[Conv1]
```

   - Conv2

```
[Conv2]vlan batch 60 80 90
Info: This operation may take a few seconds. Please wait for a moment...done.
[Conv2]

[Conv2]int g0/0/3
[Conv2-GigabitEthernet0/0/3]port trunk allow-pass vlan 10 20 60 80 90
[Conv2-GigabitEthernet0/0/3]quit
[Conv2]

[Conv2]int g0/0/4
```

```
[Conv2-GigabitEthernet0/0/4]port trunk allow-pass vlan 10 20 60 80 90
[Conv2-GigabitEthernet0/0/4]quit
[Conv2]
[Conv2]int Eth-Trunk 1
[Conv2-Eth-Trunk1]port trunk allow-pass vlan 10 20 60 80 90
[Conv2-Eth-Trunk1]quit
[Conv2]
```

- Access1

```
[Access1]vlan batch 60 80 90
Info: This operation may take a few seconds. Please wait for a moment...done.
[Access1]
[Access1]int g0/0/1
[Access1-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 20 60 80 90
[Access1-GigabitEthernet0/0/1]quit
[Access1]
[Access1]int g0/0/2
[Access1-GigabitEthernet0/0/2]port trunk allow-pass vlan 10 20 60 80 90
[Access1-GigabitEthernet0/0/2]quit
[Access1]
# Run the following commands on the interface for connecting to an AP:
[Access1]interface Ethernet0/0/22
[Access1-Ethernet0/0/22] port link-type trunk
[Access1-Ethernet0/0/22] port trunk pvid vlan 60
[Access1-Ethernet0/0/22] port trunk allow-pass vlan 60 80 90
[Access1-Ethernet0/0/22]quit
[Access1]
```

- Access2

```
[Access2]vlan batch 60 80 90
Info: This operation may take a few seconds. Please wait for a moment...done.
[Access2]
[Access2]int g0/0/1
[Access2-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 20 60 80 90
[Access2-GigabitEthernet0/0/1]quit
[Access2]
[Access2]int g0/0/2
[Access2-GigabitEthernet0/0/2]port trunk allow-pass vlan 10 20 60 80 90
[Access2-GigabitEthernet0/0/2]quit
[Access2]
# Run the following commands on the interface for connecting to an AP:
[Access2]interface Ethernet0/0/22
[Access2-Ethernet0/0/22] port link-type trunk
[Access2-Ethernet0/0/22] port trunk pvid vlan 60
[Access2-Ethernet0/0/22] port trunk allow-pass vlan 60 80 90
[Access2-Ethernet0/0/22]quit
[Access2]
```

2. Configure AP onboarding.

   Configure Conv1 as a DHCP server to ensure that users and APs can obtain IP addresses.

   - Create an address pool for APs on Conv1.

```
 [Conv1]ip pool ap
Info:It's successful to create an IP address pool.
```

```
[Conv1-ip-pool-ap]network 172.16.60.0 mask 24
[Conv1-ip-pool-ap]gateway-list 172.16.60.1
[Conv1-ip-pool-ap]option 43 sub-option 2 ip-address 10.10.10.10
[Conv1-ip-pool-ap]quit
[Conv1]
```

– Create address pools for employees and guests on Conv1.

```
[Conv1]ip pool employee
Info:It's successful to create an IP address pool.
[Conv1-ip-pool-employee]network 172.16.80.0 mask 24
[Conv1-ip-pool-employee]gateway-list 172.16.80.1
[Conv1-ip-pool-employee]dns-list 114.114.114.114
[Conv1-ip-pool-employee]quit
[Conv1]
[Conv1]ip pool guest
Info:It's successful to create an IP address pool.
[Conv1-ip-pool-guest]network 172.16.90.0 mask 24
[Conv1-ip-pool-guest]gateway-list 172.16.90.1
[Conv1-ip-pool-guest]dns-list 114.114.114.114
[Conv1-ip-pool-guest]quit
[Conv1]
```

– Enable the global address pool function on interfaces.

```
[Conv1]interface Vlanif 60
[Conv1-Vlanif60]dhcp select global
[Conv1-Vlanif60]quit
[Conv1]
[Conv1]interface Vlanif 80
[Conv1-Vlanif80]dhcp select global
[Conv1-Vlanif80]quit
[Conv1]
[Conv1]interface Vlanif 90
[Conv1-Vlanif90]dhcp select global
[Conv1-Vlanif90] quit
[Conv1]
```

– Configure the AC so that APs can go online.

Configure the IP address of the AC's source interface, and select a proper AP authentication mode to enable APs to go online.

```
# Run the following command on AP1 to obtain the MAC address of AP1:
<Huawei>display system-information
System Information
================================================
Serial Number              : 210235448310BC5A4D7B
System Time                : 2025-04-09 20:10:14
System Up time             : 1hour 30min 2sec
System Name                : Huawei
Country Code               : US
MAC Address                : 00:e0:fc:98:52:f0
Radio 0 MAC Address        : 00:00:00:00:00:00
Radio 1 MAC Address        : 00:00:00:00:00:10


# Run the following command on AP2 to obtain the MAC address of AP2:
<Huawei>display system-information
```

```
System Information
==============================================
Serial Number                   : 210235448310CD5EF006
System Time                      : 2025-04-09 20:12:33
System Up time                  : 1hour 32min 6sec
System Name                       : Huawei
Country Code                    : US
MAC Address                       : 00:e0:fc:52:15:d0
Radio 0 MAC Address          : 00:00:00:00:00:00
Radio 1 MAC Address          : 00:00:00:00:00:10
```

– Set the AC's source interface to Loopback 0.

```
[AC]capwap source interface LoopBack 0
[AC]
```

– Create the AP group Huawei on the AC.

```
[AC]wlan
[AC-wlan-view]
[AC-wlan-view]ap-group name Huawei
Info: This operation may take a few seconds. Please wait for a moment.done.
[AC-wlan-ap-group-Huawei]
[AC-wlan-ap-group-Huawei]quit
[AC-wlan-view]
```

– Set the AP authentication mode to MAC address authentication, manually add APs to the AP group, and change the AP names to AP1 and AP2.

AP1's MAC address is **00:e0:fc:98:52:f0**, which needs to be written as **00e0-fc98-52f0**.

AP2's MAC address is **00:e0:fc:52:15:d0**, which needs to be written as **00e0-fc52-15d0**.

```
[AC]wlan
[AC]wlan
[AC-wlan-view]ap auth-mode mac-auth
[AC-wlan-view]ap-mac 00e0-fc98-52f0
[AC-wlan-ap-0]ap-name AP1
[AC-wlan-ap-0]ap-group Huawei
Warning: This operation may cause AP reset. If the country code changes, it will
 clear channel, power and antenna gain configurations of the radio, Whether to c
ontinue? [Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment.. done.
[AC-wlan-ap-0]quit
[AC-wlan-view]
[AC-wlan-view]
[AC-wlan-view]ap-mac 00e0-fc52-15d0
[AC-wlan-ap-1]ap-name AP2
[AC-wlan-ap-1]ap-group Huawei
Warning: This operation may cause AP reset. If the country code changes, it will
 clear channel, power and antenna gain configurations of the radio, Whether to c
ontinue? [Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment.. done.
[AC-wlan-ap-1]quit
[AC-wlan-view]
```

– Wait for a while and run the **display ap all** command to check the AP online status. The command output shows that the two APs are online.

```
<AC1>display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
```

```
Total AP information:
nor   : normal              [2]
------------------------------------------------------------------------------
ID    MAC                  Name Group    IP                  Type                  State STA Uptime
------------------------------------------------------------------------------
0     00e0-fc98-52f0 AP1    Huawei 172.16.60.254 AP6050DN            nor     0     3M:49S
1     00e0-fc52-15d0 AP2    Huawei 172.16.60.253 AP6050DN            nor     0     3M:31S
------------------------------------------------------------------------------
Total: 2
<AC1>
```

7 Configuring WLAN Services

Configure the SSID profile, security profile, and VAP profile based on the WLAN data plan to ensure that the AP can properly release signals and STAs can connect to the AP.

1. Create SSID profiles on the AC.

   Create the SSID profiles Employee and Guest and set the SSIDs to Employee and Guest.

```
[AC1]wlan
[AC1-wlan-view]ssid-profile name Employee
[AC1-wlan-ssid-prof-Employee]ssid Employee
Info: This operation may take a few seconds, please wait.done.
[AC1-wlan-ssid-prof-Employee]quit
[AC1-wlan-view]
[AC1-wlan-view]ssid-profile name Guest
[AC1-wlan-ssid-prof-Guest]ssid Guest
Info: This operation may take a few seconds, please wait.done.
[AC1-wlan-ssid-prof-Guest]quit
[AC1-wlan-view]
```

2. Create security profiles.

   Create security profiles Employee and Guest and set security policies.

```
[AC-wlan-view]security-profile name Employee
[AC-wlan-sec-prof-Employee]security wpa2 psk pass-phrase a1234567 aes
[AC-wlan-sec-prof-Employee]quit
[AC-wlan-view]
[AC-wlan-view]security-profile name Guest
[AC-wlan-sec-prof-Guest]security open
[AC-wlan-sec-prof-Guest]quit
[AC-wlan-view]
```

3. Create VAP profiles.

   Create the VAP profiles Employee and Guest and configure them according to the WLAN data plan.

```
[AC-wlan-view]vap-profile name Employee
[AC-wlan-vap-prof-Employee]ssid-profile Employee
[AC-wlan-vap-prof-Employee]security-profile Employee
[AC-wlan-vap-prof-Employee]service-vlan vlan-id 80
[AC-wlan-vap-prof-Employee]forward-mode direct-forward
[AC-wlan-vap-prof-Employee]quit
[AC-wlan-view]
[AC-wlan-view]vap-profile name Guest
[AC-wlan-vap-prof-Guest]ssid-profile Guest
[AC-wlan-vap-prof-Guest]security-profile Guest
[AC-wlan-vap-prof-Guest]service-vlan vlan-id 90
```

```
[AC-wlan-vap-prof-Guest]forward-mode direct-forward
[AC-wlan-vap-prof-Guest]quit
[AC-wlan-view]
```

4. Create a regulatory domain profile.

   Create the regulatory domain profile Huawei and set the country code to CN.

```
[AC-wlan-view]regulatory-domain-profile name Huawei
[AC-wlan-regulate-domain-Huawei]country-code CN
Info: The current country code is same with the input country code.
[AC-wlan-regulate-domain-Huawei]quit
[AC-wlan-view]
```

5. Bind the profiles.

   Enter the view of AP group Huawei, and bind the regulatory domain profile Huawei, VAP profile Employee, and VAP profile Guest to the AP group.

```
[AC-wlan-view]
[AC-wlan-view]ap-group name Huawei
[AC-wlan-ap-group-Huawei]regulatory-domain-profile Huawei
Warning: Modifying the country code will clear channel, power and antenna gain c
onfigurations of the radio and reset the AP. Continue?[Y/N]:y
[AC-wlan-ap-group-Huawei]vap-profile Employee wlan 1 radio all
Info: This operation may take a few seconds, please wait...done.
[AC-wlan-ap-group-Huawei]
[AC-wlan-ap-group-Huawei]vap-profile Guest wlan 2 radio all
Info: This operation may take a few seconds, please wait...done.
[AC-wlan-ap-group-Huawei]
[AC-wlan-ap-group-Huawei]quit
[AC-wlan-view]
[AC-wlan-view]
```

6. Check whether the APs transmit radio signals.

   The APs transmit radio signals properly, as shown in Figure 2.

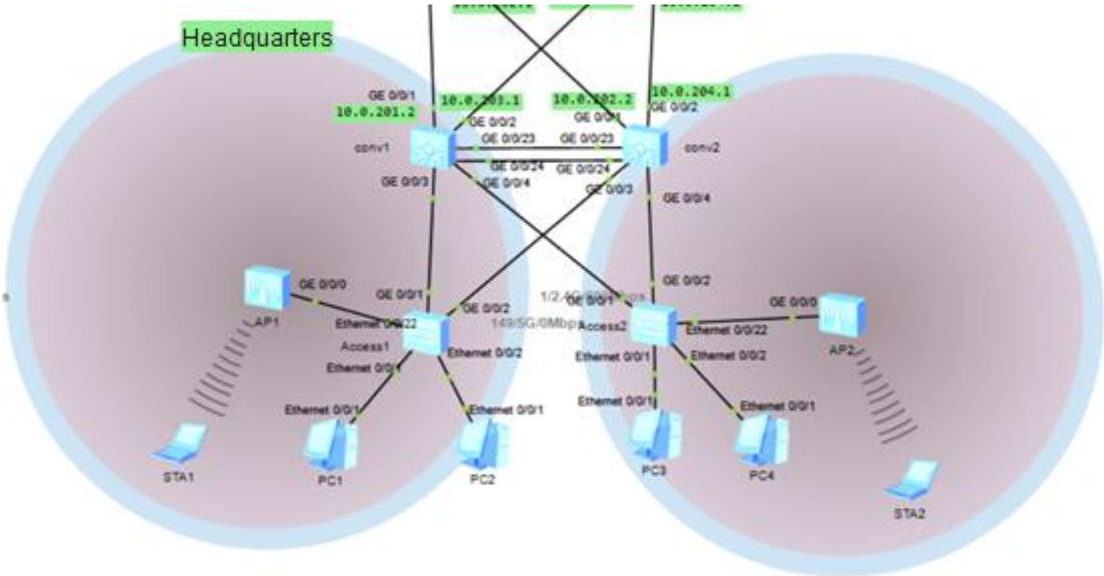

Figure 2 APs transmitting radio signals

Run the **display vap ssid Employee** command on the AC. It is found that the APs transmit dual-band signals.

```
<AC1>display vap ssid Employee
Info: This operation may take a few seconds, please wait.
WID : WLAN ID
------------------------------------------------------------------------
AP ID AP name RfID WID   BSSID            Status   Auth type   STA    SSID
------------------------------------------------------------------------
0     AP1     0    1     00E0-FC98-52F0 ON       WPA2-PSK    1      Employee
0     AP1     1    1     00E0-FC98-5300 ON       WPA2-PSK    0      Employee
1     AP2     0    1     00E0-FC52-15D0 ON       WPA2-PSK    0      Employee
1     AP2     1    1     00E0-FC52-15E0 ON       WPA2-PSK    0      Employee
------------------------------------------------------------------------
Total: 4
<AC1>
<AC1>
<AC1>display vap ssid Guest
Info: This operation may take a few seconds, please wait.
WID : WLAN ID
------------------------------------------------------------------------
AP ID AP name RfID WID   BSSID            Status   Auth type   STA    SSID
------------------------------------------------------------------------
0     AP1     0    2     00E0-FC98-52F1 ON       Open        0      Guest
0     AP1     1    2     00E0-FC98-5301 ON       Open        0      Guest
1     AP2     0    2     00E0-FC52-15D1 ON       Open        1      Guest
1     AP2     1    2     00E0-FC52-15E1 ON       Open        0      Guest
------------------------------------------------------------------------
Total: 4
<AC1>
```

8 Configuring NAT

1. Import external routes to OSPF on R1.

   Configure a default route on R1 to point to the Internet router R4. Import the default route to OSPF, and set the external route type to 1 and the cost to 20. Do not carry the **always** parameter in the command.

```
[R1]ospf 1
[R1-ospf-1]default-route-advertise type 1 cost 20
[R1-ospf-1]quit
[R1]
```

2. On PC1, ping R1 at 202.96.1.1 and R4 at 202.96.1.4.

   Because the default route is imported to OSPF on R1, PC1 can ping 202.96.1.1 of R1 but cannot ping 202.96.1.4 of R4.

   Configure Easy IP on R1.

```
[R1]acl 2000
[R1-acl-basic-2000]rule 5 permit source 10.0.0.0 0.255.255.255
[R1-acl-basic-2000]rule 10 permit source 172.16.0.0 0.0.255.255
[R1-acl-basic-2000]rule 15 permit source 192.168.0.0 0.0.255.255
[R1-acl-basic-2000]quit
[R1]
[R1]int g1/0/0
[R1-GigabitEthernet1/0/0]nat outbound 2000
[R1-GigabitEthernet1/0/0]quit
[R1]
```

3. On PC1, ping the loopback0 address 202.100.4.1 of R4.

```
PC>ping 202.100.4.1 -t

Ping 202.100.4.1: 32 data bytes, Press Ctrl_C to break
From 202.100.4.1: bytes=32 seq=1 ttl=252 time=78 ms
From 202.100.4.1: bytes=32 seq=2 ttl=252 time=94 ms
From 202.100.4.1: bytes=32 seq=3 ttl=252 time=94 ms
From 202.100.4.1: bytes=32 seq=4 ttl=252 time=93 ms
From 202.100.4.1: bytes=32 seq=5 ttl=252 time=110 ms
From 202.100.4.1: bytes=32 seq=6 ttl=252 time=109 ms

--- 202.100.4.1 ping statistics ---
  6 packet(s) transmitted
  6 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 78/96/110 ms

PC>
```

4. Run the **display nat session all** command on R1 to view the NAT mapping table.

```
[R1]display nat session all
  NAT Session Table Information:

     Protocol          : ICMP(1)
     SrcAddr   Vpn     : 172.16.10.253
     DestAddr  Vpn     : 202.100.4.1
     Type Code IcmpId  : 0   8   39636
     NAT-Info
       New SrcAddr     : 202.96.1.1
       New DestAddr    : ----
       New IcmpId      : 10249

     Protocol          : ICMP(1)
     SrcAddr   Vpn     : 172.16.10.253
     DestAddr  Vpn     : 202.100.4.1
     Type Code IcmpId  : 0   8   39635
     NAT-Info
       New SrcAddr     : 202.96.1.1
       New DestAddr    : ----
       New IcmpId      : 10248

     Protocol          : ICMP(1)
     SrcAddr   Vpn     : 172.16.10.253
     DestAddr  Vpn     : 202.100.4.1
     Type Code IcmpId  : 0   8   39634
     NAT-Info
       New SrcAddr     : 202.96.1.1
       New DestAddr    : ----
       New IcmpId      : 10247
```

9 Configuring NAT Server

Configure NAT Server on R1 to map the internal FTP server's IP address 10.0.205.10 and port number 8080 to the public IP address 122.1.2.1 and port number 80.

```
interface GigabitEthernet1/0/0
 ip address 202.96.1.1 255.255.255.0
 nat outbound 2000
#
Return
[R1]interface GigabitEthernet1/0/0
[R1-GigabitEthernet1/0/0]display this
[V200R003C00]
#
interface GigabitEthernet1/0/0
 ip address 202.96.1.1 255.255.255.0
 nat outbound 2000
#
return

[R1-GigabitEthernet1/0/0]nat server protocol tcp global 202.96.1.100 ftp inside
10.0.205.19 ftp
[R1-GigabitEthernet1/0/0]quit
[R1]quit
<R1>
```

Figure 3 shows the FTP server configuration.

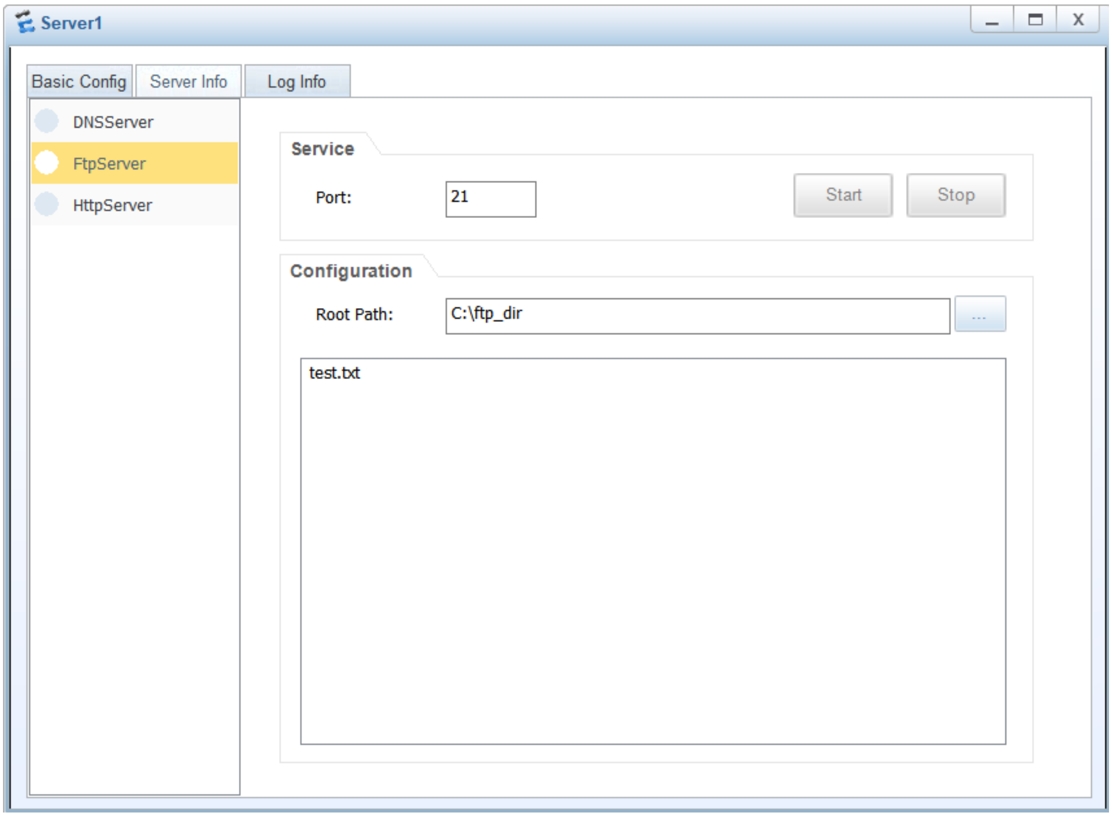

Figure 3 FTP server information

Run the **ftp** command on R1 to download the **test.txt** file from the FTP server.

```
<R1>ftp 10.0.205.10
Trying 10.0.205.10 ...

Press CTRL+K to abort
Connected to 10.0.205.10.
220 FtpServerTry FtpD for free
User(10.0.205.10:(none)):
331 Password required for   .
Enter password:
230 User   logged in , proceed

[R1-ftp]dir
200 Port command okay.
150 Opening ASCII NO-PRINT mode data connection for ls -l.
-rwxrwxrwx   1              nogroup              24 Apr 10   2025 test.txt
226 Transfer finished successfully. Data connection closed.
FTP: 68 byte(s) received in 0.180 second(s) 377.77byte(s)/sec.

[R1-ftp]get test.txt
200 Port command okay.
150 Sending test.txt (24 bytes). Mode STREAM Type BINARY
226 Transfer finished successfully. Data connection closed.
FTP: 24 byte(s) received in 0.170 second(s) 141.17byte(s)/sec.

[R1-ftp]bye
221 Goodbye.

<R1>
```

10 Configuring SSH

The SSH service configurations on routers are slightly different from those on switches. The following uses R1 and Core1 as an example. To reduce the configuration workload, the SSH user name is set to **user01** and the password to **Huawei12#$** on routers and switches.

- SSH configuration on R1

```
[R1]aaa
[R1-aaa]
[R1-aaa]local-user user01 password cipher Huawei12#$
Info: Add a new user.
[R1-aaa]
[R1-aaa]local-user user01 service-type ssh
[R1-aaa]
[R1-aaa]quit
[R1]
[R1]stelnet server enable
Info: Succeeded in starting the STELNET server.
[R1]
[R1]ssh user user01 authentication-type password
 Authentication type setted, and will be in effect next time
[R1]
[R1]user-interface vty 0 4
[R1-ui-vty0-4]authentication-mode aaa
```

```
[R1-ui-vty0-4]user privilege level 15
[R1-ui-vty0-4]protocol inbound ssh
[R1-ui-vty0-4]quit
[R1]
```

- SSH configuration on Core1

```
[core1]aaa
[core1-aaa]local-user user01 password cipher Huawei12#$
[core1-aaa]local-user user01 service-type ssh
[core1-aaa]quit
[core1]stelnet server enable
Info: The Stelnet server is already started.
[core1]ssh user user01
[core1]ssh user user01 authentication-type password
[core1]ssh user user01 service-type stelnet
[core1]user-interface vty 0 4
[core1-ui-vty0-4]authentication-mode aaa
[core1-ui-vty0-4]user privilege level 15
[core1-ui-vty0-4]protocol inbound ssh
[core1-ui-vty0-4]quit
[core1]
```

- Log in to Core1 at 10.0.2.2 from R1 using SSH.

```
[R1]stelnet 10.0.2.2
Please input the username:user01
Trying 10.0.2.2 ...
Press CTRL+K to abort
Connected to 10.0.2.2 ...
The server is not authenticated. Continue to access it? (y/n)[n]:y
Save the server's public key? (y/n)[n]:y
The server's public key will be saved with the name 10.0.2.2. Please wait...

Enter password:Huawei12#$
Info: The max number of VTY users is 5, and the number
      of current VTY users on line is 1.
      The current login time is 2025-04-10 00:44:52.
<core1>
<core1>display users
  User-Intf    Delay    Type   Network Address     AuthenStatus    AuthorcmdFlag
  0    CON 0    00:03:25                                                  no
  Username : Unspecified

+ 34   VTY 0    00:00:00   SSH     10.0.2.1                  pass            no
  Username : user01


<core1>
<core1>quit
Info: The max number of VTY users is 5, and the number
      of current VTY users on line is 0.
[R1]
```

- Log in to R1 at 10.0.2.1 from Core1 using SSH.

```
[core1]stelnet 10.0.2.1
Please input the username:user01
```

```
Trying 10.0.2.1 ...
Press CTRL+K to abort
Connected to 10.0.2.1 ...
Error: Failed to verify the server's public key.
Please run the command "ssh client first-time enable"to enable the first-time ac
cess function and try again.
[core1]ssh client first-time enable
[core1]stelnet 10.0.2.1
Please input the username: user01
Trying 10.0.2.1 ...
Press CTRL+K to abort
Connected to 10.0.2.1 ...
The server is not authenticated. Continue to access it? [Y/N] :y
Save the server's public key? [Y/N] : y
The server's public key will be saved with the name 10.0.2.1. Please wait...

Enter password:Huawei12#$
  ----------------------------------------------------------------------------

  User last login information:
  ----------------------------------------------------------------------------
  Access Type: SSH
  IP-Address : 10.0.3.1 ssh
  Time       : 2025-04-10 00:38:22-08:00
  ----------------------------------------------------------------------------
<R1>
<R1>display users
  User-Intf    Delay    Type    Network Address    AuthenStatus    AuthorcmdFlag
  0   CON 0    00:00:00                                    pass
  Username : Unspecified

+ 129 VTY 0    00:00:00   SSH    10.0.2.2                  pass
  Username : user01

<R1>quit

  Configuration console exit, please retry to log on

[core1]
```

Verification

Omitted

Quiz

1. What are the differences between SSH configurations on routers and switches?
2. In this project, what will happen after the **undo stp enable** command is run on CORE1 and CORE2?
3. What have you learned in this lab? How can the knowledge help you in your future study or work?